

URL Shortener — Tech Stack and Dependencies

Project: URL Shortener
Technology: Python, FastAPI, SQLAlchemy, SQLite, Pydantic
Location: /home/dominic/Documents/fastapiurl
Report #: 5/20

Primary Stack

Python, FastAPI, SQLAlchemy, SQLite, Pydantic

Key Dependencies

- fastapi — API framework
- sqlalchemy — ORM
- pydantic — validation
- uvicorn — ASGI server
- aiosqlite — async SQLite driver

Technology Stack Details

Framework Versions

Component	Technology	Version	Purpose
Backend	FastAPI	API server	
Database	SQLite	Persistence	
Auth	JWT	Security	
Frontend	React 19	UI	
Hosting	Render.com + GitHub Pages	Deployment	

Core Dependencies

```
# Core dependencies
fastapi>=0.115.0
sqlalchemy>=2.0.0
```

```
pydantic>=2.0.0
uvicorn[standard]>=0.30.0
```

Dependency Justification

Package	Why	Alternative Considered
fastapi	Modern ASGI framework	Flask, Django
sqlalchemy	Mature ORM with async	peewee, tortoise-orm
pydantic	Validation + serialization	marshmallow
jose	JWT handling	PyJWT

Development Dependencies

Package	Purpose
pytest	Testing framework
httpx	Async HTTP client for tests
uvicorn	ASGI server
python-dotenv	Environment variable loading

External Services

Service	Purpose	Integration Point
OpenAI	AI chat completion	/v1/chat/completions proxy
GitHub Pages	Static hosting	git push deployment

Version Constraints

- Python 3.10+ required (type hints, pattern matching)
- Pydantic v2 for improved validation performance
- SQLAlchemy 2.0+ for modern async support